



Mohammad H. Malkawi

Computer Networks & Cybersecurity Student

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Third-year Computer Networks and Cybersecurity student at Jadara University. I design systems that bring AI into physical infrastructure, and I lead student engineering programs as Vice Chair of IEEE Jadara Student Branch. Looking for internships in networking and security, remote or in Jordan.

PROJECTS

SALKA, Adaptive Traffic Control

Self-initiated

System design & research, team SYNTRA. Python, SUMO, TraCI, OpenStreetMap

- Designed a digital twin of Culture Circle in Irbid, built from OpenStreetMap data and simulated in SUMO.
- Adaptive signal control in Python over TraCI, reading queues, waiting times and occupancy, with priority for emergency vehicles.
- Reduced average vehicle waiting time by up to 15% in simulation.

C-Layer, Autonomous CO₂ Capture

Concept design

Sole designer. NVIDIA Jetson Nano, MEA-functionalized CNTs, NEMA 34 steppers, Modbus TCP, MQTT

- Designed a four-layer capture system for cement and steel stacks: pre-filtration, two MEA and carbon-nanotube absorption stages, and structural support.
- Specified robotic filter swapping in about 45 seconds with no plant shutdown, and waste-heat regeneration targeting 1.4-2.5 MJ per kg CO₂.
- Planned phased AI on the Jetson: statistical degradation models first, then LSTM prediction of remaining filter life.
- Specified protection for 870°C stacks and a spring-return bypass damper as a fail-safe on power loss.

LEADERSHIP

Vice Chair, IEEE Jadara Student Branch

Present

Chair, IEEE Robotics & Automation Society, Jadara

2025-2026

- Led a 7-person team through about 30 workshops and events.
- Organized RoboCraft on 6 January 2026, a national robotics competition on smart solutions.
- Awarded the IEEE RAS Chapter Initiative Grant in 2026 for RoboCraft. Partners included Jadara Education and IEEE Jordan Section.

110

teams registered

~300

participants

16

universities

EDUCATION

B.Sc. Computer Networks & Cybersecurity

Jadara University, Irbid

Expected January 2028

SKILLS

Networking & Security

Cisco networking, Wireshark, Kali Linux, Linux, Bash

Programming

Python, C++

Systems & AI

SUMO, TraCI, LSTM models, NVIDIA Jetson, Arduino

Tooling

Git, GitHub, Docker, OpenStreetMap data

LANGUAGES

Arabic

Native

English

Professional